Write-up

An analyst co-pilot that clarifies, checks definitions, picks the right data layer and shows its evidence.

01 What I built and why

1 Clarify — metric, period, population; asks if ambiguous

2 Verify definition — metric exists and its definition in registry + catalog

3 Route to a layer — Gold if it fits; Silver for detail; Bronze for raw audit

4 Query + evidence — read-only, bounded; one log per query, never overwritten

5 Test hypotheses — ≤3 per round, ≤3 rounds; stop when explained

6 Report — pattern ≠ cause; fixed PDF template, ≤10 pages

Read by every run, in Claude Code or Codex: AGENTS.md · SKILL.md · catalog · registry

Guardrails: read-only authorizer · row and time limits

Gold — 8 monthly tables · fast standard answers

Silver — 10 cleaned tables · detail for investigations

Bronze — 20 views over raw tables · audits

One SQLite database; layers built in one transaction with 122 reconciliation checks. Step 3 decides which layer each query reads.

- **Clarifies instead of assuming.** When the metric, period or meaning is open, it asks first and states every assumption it keeps.
- **Verifies the metric before computing it.** It confirms the metric exists and uses its approved definition and SQL from the metric registry, plus joins and known gaps from the data catalog. Unsupported requests are declined, not improvised.
- **Maps the question to a layer for speed.** Gold answers standard monthly questions directly; Silver keeps per-user detail; Bronze is only for auditing raw values. Fewer, smaller queries keep answers fast.
- **Evidence for every number and LLM-neutral by design.** Each query is logged with its SQL, layer and metric version. Definitions, rules and guardrails live in files, not prompts, so Claude Code and Codex behave the same and a lesson becomes a rule for every later run.

02 How it performed: 18 blind test questions, two rounds

	Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18
Round 1	Pass	Fail (strict rule)	Pass	Pass	Pass	Pass	Fail (strict rule)	Pass	Pass	Pass	Partial	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Round 2	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Partial	Pass	Pass	Pass	Pass	Pass	Pass	Pass

Round 1: 15 pass · 1 partial · 2 fail **Round 2: 17 · 1 · 0**

Question type	Questions	Result	What happened
Simple numbers and filters	Q01 Q05 Q08 Q17 Q18	All pass	Exact to the user and the cent
Vague or ambiguous questions	Q03 Q10 Q15 Q18	All pass	Asked before choosing a metric, period or meaning
Unanswerable question	Q16	Pass	Declined and named the data that's missing
Data-quality traps	Q12 Q13 Q14	All pass	Caught both planted issues; didn't claim to have detected a reviewed one
Comparisons and lifecycle	Q04 Q06 Q09	All pass	Right time basis; groups reconcile to the total
Cohort conversion	Q07	Fail → Pass	Used April, the month the subscriber base forms, as a normal baseline. A new skill rule fixed it
Funnel drop-off	Q11	Partial	Right answer and gaps, but read April's start-up as a funnel decline; the new rule didn't trigger here
GM investigation	Q02	Fail → Pass	v1: headline numbers and story right, but some conclusions stated more firmly than the evidence allowed. After review, three certainty rules joined the skill; v2 and a blind rerun on the updated skill both passed

Scored against values computed beforehand from the raw tables. Correct numbers alone don't pass: a silent choice of definition, a missing caveat or an overstated conclusion fails. Round 2: Q02 was revised after review and rerun blind; Q07 and Q11 were rerun, with Q01 and Q09 to check nothing broke; other results carry over. **Hardest:** comparisons that need context the data doesn't announce, such as a start-up month.

03 Running it for a 20-person product team

Time to answer (observed)	**Simple** (MAU by month): 2–3 min	**Normal** (refund impact): 5–6 min	**Complex** (MAU up, revenue flat): 10–12 min

- **How the team uses it:** analysts and PMs ask in plain English from Claude Code or Codex (later Slack or a web page), all on one shared repository, so definitions and guardrails are the same for every person and model.
- **Set-up:** build a data catalog and metric registry for the team's warehouse, with the analytics lead approving each definition, then the Gold and Silver layers the agent routes to.
- **Cost:** mostly model tokens, which grow with question complexity and how much the agent reads; queries cost next to nothing. Not measured here: log tokens per question first, then budget by question type. Keep it down by answering from the smallest layer that fits, pre-building common questions into Gold and giving simple questions a short answer, which also cuts latency.
- **Guardrails:** data is only ever read (read-only role plus the query tool's check), with row and time limits; every figure traceable to its evidence log; sensitive columns kept out of the catalog; unclear questions get a question back.
- **Freshness:** real data keeps arriving, unlike this static dataset. A freshness tracker (last load per source and layer) is checked first; the agent warns or stops when data is stale.
- **Quality over time:** a test set of questions with known answers (these 18, growing from real ones), rerun on every change to the skill, definitions or data; track pass rate, clarification rate, query errors and user ratings; turn failures into rules.

04 With more time

- Run the same questions in both runtimes for a direct Claude vs Codex comparison, plus held-out questions.
- Measure tokens and time per question to put a real cost on each question type.
- A short-answer mode for simple questions; a helper that drafts the catalog for a new database; a Slack or web front end.

Note on tools. Claude Code (Claude Opus 5.5, medium) built the framework, ran the GM investigation and computed the answer key. Codex (GPT Terra 6.1, medium) ran the evaluation, so the agent was tested in a runtime that didn't build it. ChatGPT (Terra 5.6, later 6.1, medium) reviewed the GM report and evaluation decisions independently.